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Myungjin Lee (she/her)

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Interests

Graph Representation Learning, Molecular Property Prediction, Drug Repurposing, Explainable AI (XAI)

Education

M.E. in Computer Science and Artificial Intelligence Mar 2024 – Aug 2026
Dongguk University, Seoul, Republic of Korea
- Advisor: Dr. Sangsoo Lim
- Master's thesis: *Graph Substructure-Based Junction Tree Modeling for Drug ADMET Prediction*
- Major: AI in Healthcare and Medicine
- GPA 4.04/4.5

B.E. in Industrial & Management Engineering Mar 2023 – Feb 2024
(Advanced Major Program)
Induk University, Seoul, Republic of Korea
- GPA 4.43/4.5; Valedictorian

A.E. in Industrial & Management Engineering Mar 2020 – Feb 2023
Induk University, Seoul, Republic of Korea
- GPA 4.41/4.5 (Major: 4.42/4.5); Valedictorian

Research Experience

Dongguk Univ. PRISM Lab, Seoul, Republic of Korea Jan 2024 – Present
- *Position:* Master's Researcher (Sep 2026 – Present); Graduate Researcher (Jan 2024 – Aug 2026)
- Advisor: Dr. Sangsoo Lim
- *Role:* Molecular graph representation learning and explainable AI for drug discovery — interpretable ADMET prediction from molecular substructures (first-author SJoINT; co-author MAGNET); a drug-repurposing model developed and registered on the national bio-data platform (KBDI); and bulk/single-cell RNA-seq analysis of cellular senescence and aging-related MASLD with Mayo Clinic collaborators.

Manuscripts

(Under review) **Lee M.**, Mo J., Kang M. & Lim S.*, "SJoINT: Substructure-Driven Junction Tree for Interpretable Molecular Property Prediction"
- Dual-view learning of atom-level graphs and substructure-level junction trees via structure-constrained bidirectional cross-attention with augmentation-free contrastive pretraining (ZINC250k); evaluated on MoleculeNet ADMET and MoleculeACE activity-cliff tasks.

(Under review) Mo J., **Lee M.**, Lee S. & Lim S.*, "MAGNET: Cross-view structural consensus for interpretable molecular property prediction"
- Molecular graph learning framework integrating BRICS, junction-tree, and Murcko scaffold decompositions through explicit fragment correspondences in a unified meta-graph, with multi-objective self-supervised pretraining; best performance on seven of ten molecular property benchmarks.

Manuscripts (cont.)

(In preparation) Lee G., Byeon S.K., Olaleye O.E., Han Y., **Lee M.**, Vizioli M.G., Ogrodnik M., Pirius N., Sales Gomez L., Victorelli S., Khosla S., Pandey A., Passos J., Lim S. & Jurk D.*, "Lipid abundance and lysophospholipid signaling regulate the senescence-associated secretory phenotype"

- Lipid accumulation and lysophosphatidylcholine signaling drive the SASP through LPA receptor-dependent p38/JNK MAPK activation; fatty acid synthase inhibition suppresses senescence-associated inflammation and improves frailty in aged mice.
- *Role:* transcriptomic analysis of public, in vitro, and in vivo senescence data (differential expression, gene-set enrichment); collaboration with Mayo Clinic.

Conference Papers

Lee M., Mo J. & Lim S.*, "Structure-Constrained Bidirectional Cross-Attention for Self-Supervised Molecular Representation Learning", *Proceedings of the Korea Computer Congress (KCC)*, Korean Institute of Information Scientists and Engineers (KIISE), pp. 291–293, 2026

Poster Presentations

BIOINFO/GIW ISCB-Asia 2026, Yonsei University, Seoul, Korea — Nov 2026

- *Cross-attentive Integration of Atom-level and Junction-tree Representations for Molecular Property Prediction*

Korea Computer Congress (KCC) 2026, ICC Jeju, Korea — Jun 2026

- *Structure-Constrained Bidirectional Cross-Attention for Self-Supervised Molecular Representation Learning*

BIOINFO 2025, Suwon Convention Center, Korea — Oct 2025

- *Weighted Junction-Tree Nodes for Enhanced Interpretability in ADMET Tasks*

Grants

석사과정생 연구장려금지원사업 — PI — Sep 2024 – Aug 2025

- *Ministry of Education (MOE) / National Research Foundation of Korea (NRF).* Individual research grant on functional-group junction-tree-based multi-task molecular graph representation learning.
  (Advisor: Dr. Sangsoo Lim)

Research Projects *(Participating researcher)*

멀티모달 AI 기반 노화연관 지방간질환(MASLD) 표적 발굴 및 약물 검증 *(RS-2025-00560523)* — Mar 2025 – Feb 2026

PI: Dr. Sangsoo Lim (Dongguk University);
Collaborator: Dr. Gung Lee (Mayo Clinic, USA).

- *Role:* Curated public bulk and single-cell RNA-seq datasets into an integrated database for aging-related MASLD and defined stages of disease progression; analyzed stage-wise gene expression changes with functional enrichment (GSEA: GO, KEGG, Hallmark) and cellular-senescence gene-set analysis (SenMayo, SHGS).

개방형 AI 신약개발·데이터 분석 플랫폼 개발센터 — Jul 2025 – Aug 2026

(RS-2025-18732993)
PI: Dr. Minho Lee (Dongguk University)

- *Role:* **(Drug repurposing)** built a standardized PrimeKG-based benchmark to evaluate five models (TxGNN, DREAMWalk, DRHGCN, AMDGT, SVGA) across 190 diseases / 3,379 drugs; developed DC-VGAE and a performance-based ensemble. **(ADMET)** benchmarked 13 AI models for physicochemical-property and toxicity prediction on nine MoleculeNet datasets; developed an interpretable ADMET model using junction-tree / Murcko / BRICS substructures with cross-attention. (Deployment) packaged the drug-repurposing model and the ADMET benchmark and registered them on the national bio-data platform (KBDI).

Challenges

2024 Samsung AI Challenge — Machine Learning Force Field Aug 2024
- Samsung Electronics SAIT · DACON
- **Top 10% (11th)**
- Built MLFF models predicting atomic energies and forces with uncertainty quantification for semiconductor-material simulation (team DAI).

DREAM Olfactory Mixtures Prediction Challenge Aug 2024
- International DREAM Challenge, Sage Bionetworks
- Developed a graph neural network to predict the olfactory similarity of molecular mixtures.

Honors and Awards

- **Highest Academic Achievement Award** 2023

Induk University, Korea

- **Merit-based Academic Scholarship** 2020 – 2023
  — awarded every semester

Induk University, Korea (incl. GRAPE Talent Scholarship)

Patents and SW Registration

Software Registration
- (No. C-2025-040207) An Embedding Method via Multi-view Graph Integration of Molecular Structures, Korea Copyright Commission, 2025
- (No. C-2024-042597) Deep Learning Model for Similarity Prediction of Graph-based Mixtures, Korea Copyright Commission, 2024

Korean Patent Application (pending)
- (No. 10-2025-0003146) System and Method for Predicting Olfactory Characteristics of Odor Mixture Data, 2025

Technical Skills

Programming	Python, R (basic), Java, C#, JavaScript, SQL
Deep Learning	PyTorch, PyTorch Geometric, Hugging Face Transformers, scikit-learn
Data Analysis & Visualization	pandas, NumPy, anndata, NetworkX, seaborn, matplotlib
Cheminformatics	RDKit, DeepChem, TorchDrug, Chemprop
Bioinformatics	Scanpy, Seurat, bulk RNA-seq and scRNA-seq analysis, differential expression (DESeq2 / PyDESeq2, MAST), GSEA / ssGSEA, GO enrichment
Tools & Databases	Git, Linux, Docker, MySQL

Languages

Fluent in **English** and Native in **Korean**
- TOEIC 860 (LC 440 / RC 420)
- OPIc (English) IH